

Spanish Ciphers before Accession of King Ferdinand: 1470-1479

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The nineteenth-century historian Gustav Bergenroth said cipher was introduced to Spain by Miguel Pérez de Almazán, Secretary of King Ferdinand. The present article describes several ciphers Ferdinand used before his accession in his letters to his father, King John II of Aragon in the 1470s.

1. Introduction

Generally, it is not easy to specify the person who started use of cipher in a country. For Spain, the nineteenth-century historian Gustave Bergenroth said cipher was introduced to Spain by Miguel Pérez de Almazán, Secretary of King Ferdinand (Bergenroth p.xii; Real Academia de la Historia). As of 1860, he said "I have seen no despatches written in cipher of an earlier date than the year 1480, and even then the ciphers were for about fifteen years of a rather simple and primitive character." (Cartwright, p.79)

It must be remembered, however, that his so-called "simple and primitive" ciphers can be a code with 500 entries for names of persons and countries. One cipher used in a letter of 26 May 1491 of Queen Isabella in writing to Dr. Puebla in London appears to have had entries up to at least 439, including not only names but also verbs, adjectives, and other functional words (Bergenroth p.35). A code of this size could pass as a seventeenth-century cipher.

Moreover, despite Bergenroth's statement in 1860, ciphers were already used in Ferdinand's letters to his father in the 1470s, the fact already pointed out decades ago by Walsh (p.490).

The present article describes several ciphers Ferdinand used before his accession in letters to his father, King John II of Aragon, preserved in the National Library of Spain (BNE). The letters used in this article are all accompanied with decipherment in the manuscript, which allowed me to reconstruct four ciphers.¹

2. Ciphers

2.1 Ferdinand-John II of Aragon Cipher (1470)

Source:

Prince Ferdinand to his father, John II of Aragon, Dueñas, 12 November 1470

MSS/20211/56, <http://bdh.bne.es/bnearch/detalle/bdh0000186631>

Ferdinand of Aragon (1452-1516) and Isabella of Castile (1451-1504), later called the Catholic Monarchs, married in October 1469 and held court at Dueñas (Prescott, p.130). They were blessed with a daughter on 2 October 1470 (the "infanta" is mentioned in the letter). However, Isabella's brother,

¹ The letter of 12 November 1470 is quoted by Arturo Quirantes, who says the interlined decipherment was written by Bergenroth. Indeed, it appears to be in pencil and evidently belong to a later century. The materials from the National Library of Spain were not included in Bergenroth's scope for vol. 1 (dealing with the years of Henry VII of England) but only for vol. 2 (Henry VIII), as explained in the beginning of the volumes.

King Henry IV of Castile, was opposed to the marriage and in October 1470, Henry declared that Isabella forfeited whatever claims she had by marrying without his consent and the King further swore the legitimacy of his daughter Joanna, whom he announced to be his successor (Prescott p.130).

The cipher used in this letter of November 1470 can be reconstructed as follows.²

a	b	c	d	e	f	g	h	i	l	m	n	o	p	q	r	s	t	u	x	y	z	?
⊖	4	h	9	⊖	⊖	⊖	⊖	5	7	9	11	+	⊖	1/2	17	19	20					5 7 11
⊖			⊖				7	8			12		9	10	18	8	21					8
⊖			4								13		⊖				22					⊖
⊖																	4					6
⊖																						
ba	be	bi	bo	bu		la	le	li	lo	lu		sa	se	si	so	su		ll?	37			
⊖		⊖				⊖		⊖	⊖	⊖		⊖	⊖	⊖								
ca	ce	ci	co	cu		ma	me	mi	mo	mu		ta	te	ti	to	tu					90	
⊖	⊖	⊖	⊖			⊖	⊖	⊖	⊖		⊖	⊖	⊖	⊖	⊖						por	91
																					que	92
da	de	di	do	du		na	ne	ni	no	nu		xa	xe	xi	xo	xu					93	
⊖	⊖	⊖	⊖	⊖		⊖		⊖	⊖	⊖		⊖									del	97
fa	fe	fi	fo	fu		pa	pe	pi	po	pu		ya	ye	yi	yo	yu						
⊖		⊖				⊖				⊖												
ga	ge	gi	go	gu		ra	re	ri	ro	ru		za	ze	zi	zo	zu						
		⊖	⊖			⊖		⊖				⊖										

Cipher Used in Letters of Ferdinand the Catholic before Accession (1470)

Fig.1

Far from being "simple and primitive", this is quite developed a system for the age. Not only the substitution alphabet has homophones and short words are provided for, apparently the full set of syllables (consonant-vowel combinations) are given their own symbols. Later, Spanish ciphers employed a vowel indicator system, whereby syllables are systematically represented by combinations of base symbols and diacritics (see, for example, S. Tomokiyo (2)). In contrast, with this cipher, the symbols representing syllables are just arbitrary symbols (though some tendency for regularity can be seen). From a cryptographic viewpoint, this provides higher security than later vowel indicator ciphers, albeit at the expense of convenience for the correspondents. As far as known to

² Some symbols, especially those like φ and 7 as well as extra symbols for syllables, are not certain. In addition, the distinction between i and y as well as between b and u is not certain. (Apparently, those symbols are used interchangeably.) In symbols for L and R, the Arabic numeral "5" has an old style, as is the case with other specimens.

me, a cipher of this kind is not found until what I provisionally call a Figueroa-Leyva Cipher (1533) (Tomokiyo (2)).

2.2 Ferdinand-John II of Aragon Cipher (1473)

Source:

Prince Ferdinand to his father, John II of Aragon, Hostabrich, 5 June 1473

MSS/20211/64, <http://bdh.bne.es/bnearch/detalle/bdh0000186643>

Aragon during the reign of John II was filled with internal and external troubles. Young Ferdinand was often recalled to Aragon by his father for help. No sooner had the Catalan Civil War ended with the surrender of Barcelona in 1472 (Prescott p.101) than King John began a campaign to regain border provinces of Roussillon and Cerdagne from the French King and Ferdinand led Castilians to his father's relief in Perpignan (capital of Roussillon) north of the Pyrenees (Prescott p.132-133). ("Hostabrich" probably refers to Hostalric, a village near Barcelona.) A truce was agreed in July 1473 and a treaty was concluded in September 1473 (Wikipedia).

The reconstructed cipher is a simple substitution cipher with homophones.

a	b	c	d	e	f	g	h	i	l	m	n	o	p	q	r	s	t	u	x	y	z
o	3	p	4	i		9	8	h	+	a	2f	o		re	b	f	m	offo		b	
r		9		8				o	++		9o							9			
2				8							P							+++			
o				a					oo												
Cipher Used in Letters of Ferdinand the Catholic before Accession (1473)																					

Fig. 2

2.3 Ferdinand-John II of Aragon Cipher (1474-1476)

Sources:

Prince Ferdinand to his father, John II of Aragon, Zaragoza, 3 November 1474

MSS/20211/73, <http://bdh.bne.es/bnearch/detalle/bdh0000186628>

Prince Ferdinand to his father, John II of Aragon, Bilbao, 5 August 1476

MSS/20211/98, <http://bdh.bne.es/bnearch/detalle/bdh0000186664>

The French conquered Roussillon in the summer of 1474 and Ferdinand was once again summoned by his father to his assistance and went to Zaragoza (Saragossa) to obtain aid from the estates of Aragon (Prescott p.137-138). Despite the prince's efforts, Perpignan fell in March 1475 (Wikipedia).

In December 1474, Henry IV of Castile died, and Ferdinand hastened back to Castile (Prescott p.145). He now had to face a rival claimant to the Castilian throne, supported by Portugal, opposing the succession of Isabella in the War of the Castilian Succession. Through a tactically inconclusive Battle of Toro in March 1476, Isabella's succession was secured (Prescott p.153-157). After the victory of Toro, Ferdinand could divert his attention to the western end on the border with France and went to Biscay to discuss with his father some measures for the pacification of Navarre (Prescott p.157-158). The

second specimen above is from this period, from which the cipher has been reconstructed as shown below.

This is a simple substitution cipher with homophones, similar to the Ferdinand-John II of Aragon Cipher (1473) above.

a	b	c	d	e	f	g	h	i	l	m	n	o	p	q	r	s	t	u	x	y	z	nulls	que
																							
																							
																							
																							

Cipher Used in Letters of Ferdinand the Catholic before Accession (1476)

Cipher Used in Letters of Ferdinand the Catholic before Accession (1476)

Fig. 3

2.4 Ferdinand-John II of Aragon Cipher (1476-1479)

Sources:

Prince Ferdinand to his father, John II of Aragon, Madrigal, 30 April 1476

MSS/20211/94, <http://bdh.bne.es/bnearch/detalle/bdh0000186601>

Prince Ferdinand to his father, John II of Aragon, Medina del Campo, 6 July 1477

MSS/20211/105, <http://bdh.bne.es/bnearch/detalle/bdh0000186630>

Prince Ferdinand to his father, John II of Aragon, 1477

MSS/20211/109, <http://bdh.bne.es/bnearch/detalle/bdh0000186570>

Prince Ferdinand to his father, John II of Aragon, Madrid, 9 April 1478 *mainly used in the reconstruction

MSS/20211/114, <http://bdh.bne.es/bnearch/detalle/bdh0000186666>

Prince Ferdinand to his father, John II of Aragon, Madrid, 18 April 1478

MSS/20211/115, <http://bdh.bne.es/bnearch/detalle/bdh0000186576>

Prince Ferdinand to his father, John II of Aragon, Madrid, 18 April 1478

MSS/20211/116, <http://bdh.bne.es/bnearch/detalle/bdh0000186612>

Prince Ferdinand to his father, John II of Aragon, Cordoba, 4 November 1478

MSS/20211/123, <http://bdh.bne.es/bnearch/detalle/bdh0000186627>

Prince Ferdinand to his father, John II of Aragon, Trujillo, 4 December 1478

MSS/20211/126, <http://bdh.bne.es/bnearch/detalle/bdh0000186569>

Prince Ferdinand to his father, John II of Aragon, Trujillo, 22 January 1479

MSS/20211/128, <http://bdh.bne.es/bnearch/detalle/bdh0000186571>

After the victory of Toro, Ferdinand, joined by Isabella at Medina del Campo, went to Madrigal and assembled a cortes of three estates, securing allegiance to the King and Queen as well as to the Infanta as heiress (Walsh p.177-178).

Ferdinand and Isabella stayed on occasions in Trujillo in Estremadura after retaking it, where Isabella signed a peace treaty with Portugal in September 1479. Ferdinand was then back in Aragon and signed a separate document (Edwards). He was now King of Aragon by the decease of his father John II on 20 January 1479.

Of the letters listed above, Cipher (1476-1479) was first used in Madrigal, where Ferdinand stayed with Isabella. It is yet to be found out whether this was a development of Cipher (1470) or was brought by Isabella from the Castilian court.

Obviously, Cipher (1470) and Cipher (1476-1479), with their full representation of syllables, could not have been the first Spanish ciphers and must have been preceded by decades of history of development, which is yet to be explored.⁴

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William Thomas Walsh, *Isabella of Spain*, first published in 1931, p.490

<https://archive.org/details/TheLastCrusaderIsabellaOfWalshWilliamThomas5474/page/n489>

⁴ A few days after uploading this article, I found Spanish ciphers of the 1470s and earlier have been described in footnotes 2 and 5 in Galende Diaz (1993-1994), “La escritura cifrada durante el reinado de los Reyes Catolicos y Carlos V” (<http://digibug.ugr.es/bitstream/handle/10481/30410/CEM-018-019-Art%C3%ADculo-009.pdf?sequence=1>). Specifically, of Ferdinand’s letters treated herein, those of 12 November 1470 and 3 November 1473 are mentioned, along with other letters between other correspondents.